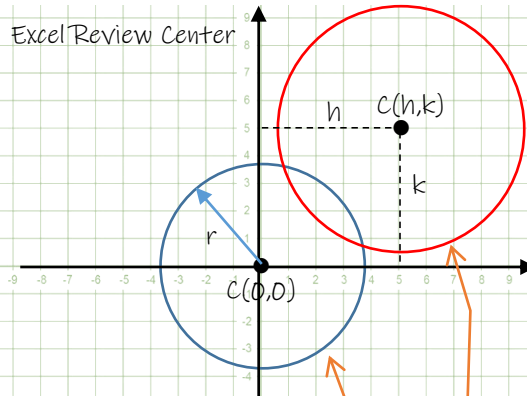




Circle

A locus of a point that which moves so that it is equidistant from a fixed point called center.



General Equation

$$x^2 + y^2 + Dx + Ey + F = 0$$

If D and E = 0, center is at origin

Standard Equations

$$C(0,0)$$

$$C(h,k)$$

$$x^2 + y^2 = r^2$$

$$(x-h)^2 + (y-k)^2 = r^2$$

When the equation given is general equation rather than standard equation, the center (h,k) of the circle and its radius (r) can be determine by converting the general equation to standard using the process known as completing the square. Or using the following formulas:

General Equation $Ax^2 + Cy^2 + Dx + Ey + F = 0$

Center (h, k)

$$h = \frac{-D}{2A}$$

$$k = \frac{-E}{2A}$$

Radius, r

$$r = \sqrt{\frac{D^2 + E^2 - 4AF}{4A^2}}$$

Parabola Continuation...

Additional formulas for h, k and a:
For axis horizontal: $Cy^2 + Dx + Ey + F = 0$

$$h = \frac{E^2 - 4CF}{4CD}$$

$$k = \frac{-E}{2C}$$

$$a = \frac{-D}{4C}$$

For axis vertical: $Ax^2 + Dx + Ey + F = 0$

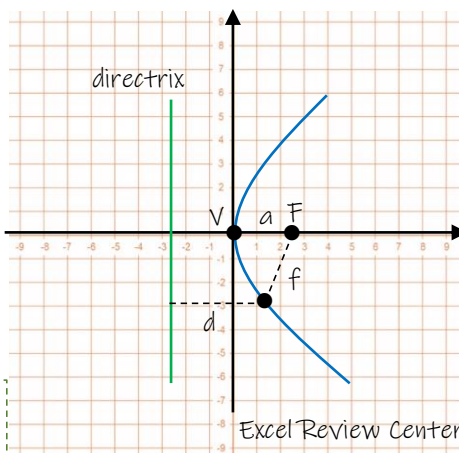
$$h = \frac{-D}{2A}$$

$$k = \frac{D^2 - 4AF}{4AE}$$

$$a = \frac{-E}{4A}$$

Parabola

A locus of a point which moves so that it is always equidistant to a fixed point called focus and it a fixed straight line called directrix.



General Equation

Axis parallel to y-axis

$$Ax^2 + Dx + Ey + F = 0$$

Axis parallel to x-axis

$$Cy^2 + Dx + Ey + F = 0$$

Standard Equation

Vertex at origin (0,0).

Axis parallel to x-axis

Open to right

$$y^2 = 4ax$$

Open to Left

$$y^2 = -4ax$$

Axis parallel to y-axis

Open Up

$$x^2 = 4ay$$

Open Down

$$x^2 = -4ay$$

Vertex at (h,k).

Axis parallel to x-axis

Open to the right

$$(y-k)^2 = 4a(x-h)$$

Open to the left

$$(y-k)^2 = -4a(x-h)$$

Axis parallel to y-axis

Open Up

$$(x-h)^2 = 4a(y-k)$$

Open Down

$$(x-h)^2 = -4a(y-k)$$

LATUS RECTUM
- line that passes through the focus and perpendicular to the axis of the conic

$$LR = 4a$$

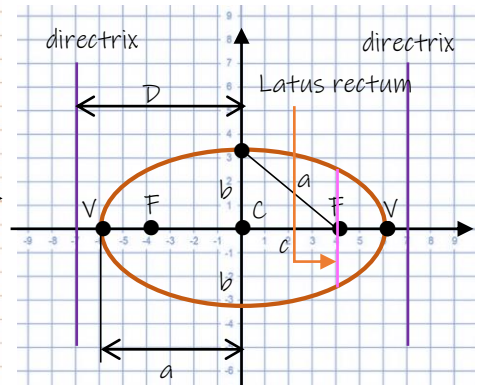
ECCENTRICITY
- the ratio of the distance to the focus to the distance to the directrix

$$e = 1$$

Ellipse

Excel Review Center

A locus of a point which moves so that the sum of its distance to the fixed points (foci) is constant and is equal to the length of the major axis (2a).



2a = major axis 2b = minor axis

General Equation

$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

If D and E = 0, center is at origin otherwise it is at h,k

Standard Equation

Center at origin (0,0).

Major axis horizontal

Major axis vertical

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

Center at origin (h,k).

Major axis horizontal

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Major axis vertical

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

LATUS RECTUM

$$LR = \frac{2b^2}{a}$$

ECCENTRICITY

$$e = \frac{a}{D} = \frac{c}{a} < 1.0$$

When the equation given is a general equation rather than standard equation, the center (h,k) of an ellipse and its focal length "c" can be calculated by converting the general equation to standard using the process known as completing the square. The following formulas can be obtained:

General equation:

$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

$$h = \frac{-D}{2A}$$

$$k = \frac{-E}{2C}$$

$$c = \sqrt{a^2 - b^2}$$